GOOGLE AI BUILDER CUP 2026 · FUTURE OF WORK & ENTERPRISE PRODUCTIVITY

KnowBridge
Connecting knowledge to confident decisions

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TEAM LEADER	**Alfian Hidayatulløh**
PROBLEM STATEMENT	Google Workspace SMEs hold project commitments across scattered Docs and Sheets — customer targets in one file, QA forecasts in another, security sign-off in a third. Nothing reconciles **baseline vs forecast vs commitment vs approved decision**, so teams risk promising dates the evidence does not support.

All organizations, people and records in this deck are fictional synthetic data. Results are reported as measured.

BRIEF ABOUT THE IDEA

An evidence catalog that answers with sources, dates and honest gaps

KnowBridge is a grounded search and decision-support tool for Google Workspace SMEs. It imports a bounded set of files, keeps the original text and record dates attached to every piece of evidence, and answers questions with **literal, click-through citations** — reconciling conflicting documents instead of hiding them.

The workflow

Connect → **Investigate** → **Reconcile** → **Decide** — inside the tools teams already use, with every answer traceable to source text.

The output

A structured **Decision Brief**: options, risks, unknowns and conditional recommendations — for a human to verify. Never an autonomous decision.

The discipline

When evidence is absent, the answer is **"insufficient evidence"**. Fabricated excerpts are rejected server-side; a later forecast never silently erases an earlier commitment.

The proof

Deployed on Google Cloud with a deterministic synthetic corpus and a 12-case live benchmark. Failures are reported as measured, including service errors.

OPPORTUNITIES

Why KnowBridge is different — and how it solves the problem

- **How is it different from existing ideas?** Enterprise search returns passages. KnowBridge **types** evidence (baseline / forecast / commitment / approved decision), preserves each document's recorded date, and surfaces conflicts across files instead of flattening them into one answer.
- **How does it solve the problem?** One question replaces file archaeology: grounded retrieval → literal citation verification → temporal and conflict analysis → a decision brief with explicit unknowns. Grounding gates run **before and after** the model; if they fail, the answer is withheld (fail-closed).
- **USP of the proposed solution** Temporally aware, explainable decision support: a revised forecast does not erase a customer commitment, "pending" is never shown as "approved", and every quoted excerpt must literally exist in an authorized source — verified server-side, not just prompted.

Market framing is stated as a testable hypothesis (Singapore services SMEs on Google Workspace) — not a validated market claim.

LIST OF FEATURES OFFERED BY THE SOLUTION

Features

- **Bounded Workspace import** — Google Picker + OAuth; registered fictional files only; Sheets import first tab.
- **Source catalog** — every record with stable ID, owner, recorded date and full original text.
- **Grounded Ask** — answers with literal, click-through citations to exact excerpts.
- **Abstention** — absent evidence is reported as "insufficient evidence", never invented.
- **Temporal & conflict analysis** — baseline vs forecast vs commitment vs approved decision, typed.
- **Decision Brief** — options, risks, unknowns, conditional next steps for human review.
- **Save & review briefs** — authorized member sessions; revocation-checked Firebase ID tokens on mutations.
- **Effective-date discipline** — document effective date is distinguished from any Drive updated timestamp.
- **Tenant isolation** — retrieval is restricted to authorized tenant/user documents before the LLM is called.
- **Fail-closed integrity gates** — fabricated excerpts, missing citations or coverage gaps block the answer.

PROCESS FLOW DIAGRAM

Use-case flow: connect → investigate → reconcile → decide

1 · Connect

Member selects registered files via Google Picker (OAuth). Only authorized, tenant-pinned sources are imported and indexed.

2 · Investigate

Ask a question. Tenant-scoped retrieval + grounded answer with literal excerpts and source links. Absent evidence → abstain.

3 · Reconcile

Temporal & conflict analysis across sources: what is baseline, forecast, commitment, approved decision — and since when.

4 · Decide

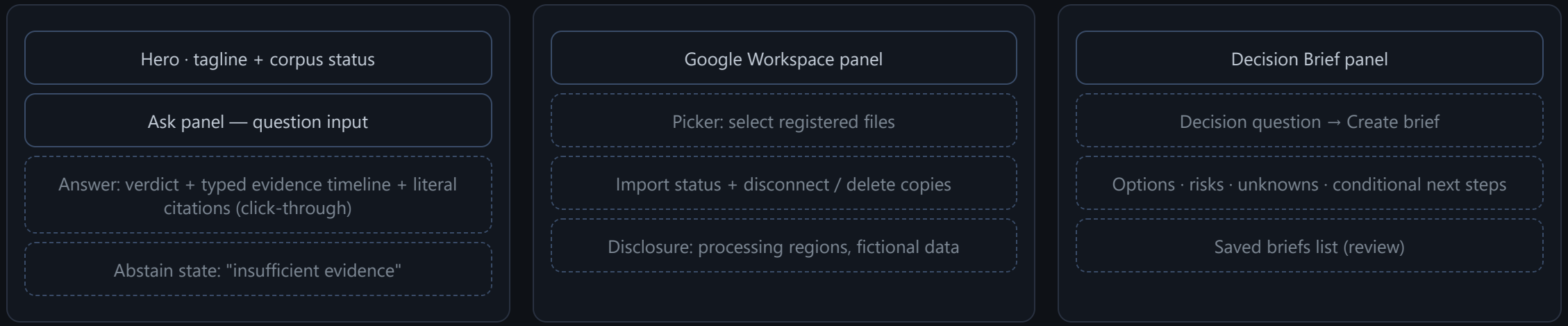
Generate a Decision Brief (options, risks, unknowns) for human review. Saved for review; never auto-approved.

Grounding gates on every answer: tenant-scoped retrieval → literal citation verification → decision-coverage check → semantic review → fail closed (502/503) if any gate fails.

Actors: SME project lead / delivery owner (asker) · authorized Workspace member (imports, reviews briefs) · human decision-maker (approves — the app never does).

WIREFRAMES / MOCK DIAGRAMS (OPTIONAL)

Interface structure — wireframe



Below the panels: **Source catalog** — list of imported records on the left, full record view on the right (stable ID, owner, recorded date, original source text).

Wireframes are structural sketches; the following slides show the actual deployed prototype.

ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION

Google Cloud architecture — fail-closed grounding

Google Workspace

Docs / Sheets (fictional). Google Picker + OAuth for selected-file import. Document effective dates travel with the content.

▸

Cloud Run · asia-southeast1

Next.js / Node 22 API. Non-root container, Secret Manager, CSP. Grounding gates; revocation-checked Firebase ID tokens on mutations; fail-closed 502/503.

▸

Gemini + Firestore

gemini-3.8-flash (global endpoint) for answers & briefs; **gemini-embedding-001** 768-d vectors (us-central1); Firestore vector retrieval, tenant-scoped.

Answer pipeline

Tenant-scoped retrieval → model call → literal citation verification → decision-coverage check → semantic review → ship or fail closed.

Security basics

Non-root container · Secret Manager secrets · revocation-checked tokens on writes · CSP · synthetic-tenant pinning · cross-tenant chunks never reach the LLM.

Honest disclosure: model processing is not Singapore-only (Gemini global endpoint; embeddings US). Real-data import is prohibited.

TECHNOLOGIES USED IN THE SOLUTION

Stack

Application

Next.js 15 · React 19 | TypeScript | Node 22 | Zod validation

Single deployable web app + API routes; strict schema validation at the boundaries.

Google AI & data

Gemini (gemini-3.8-flash) | gemini-embedding-001 | Firestore vector search

Grounded generation with literal-citation verification; 768-d embeddings for retrieval.

Google Cloud & Workspace

Cloud Run (asia-southeast1) | Secret Manager | Docker | Google Picker + OAuth | Firebase Auth (ID tokens)

Service identity over hard-coded keys; revocation-checked tokens on mutations.

Quality & evaluation

Vitest | Playwright | axe-core accessibility | Python seed generator

256 unit/integration tests, e2e flows, live adversarial semantic regressions, deterministic synthetic dataset.

ESTIMATED IMPLEMENTATION COST (OPTIONAL)

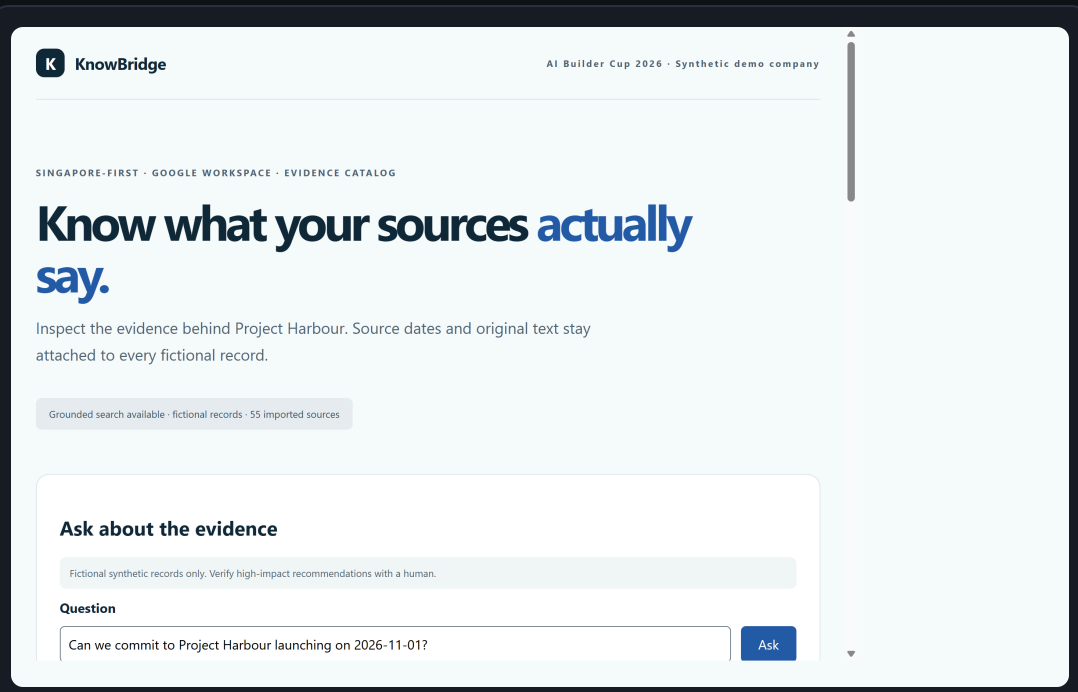
Running cost — rough order-of-magnitude estimates

Component	Basis	Estimated monthly cost (hackathon / demo scale)
Cloud Run (asia-southeast1)	1 vCPU · scale-to-zero · demo + evaluation traffic	≈ S$5 – 15
Gemini API (flash + embeddings)	Usage-based; demo queries + 12-case benchmark runs are the main driver	≈ S$10 – 25
Firestore (vector retrieval)	Small synthetic corpus (168 docs / 24 sheets); mostly free tier	≈ S$0 – 5
Secret Manager · Artifact Registry	Secret versions + container images	≈ S$1 – 3
Frontend hosting	Served from the same Cloud Run service	≈ S$0
Total		**≈ S$20 – 50 / month**

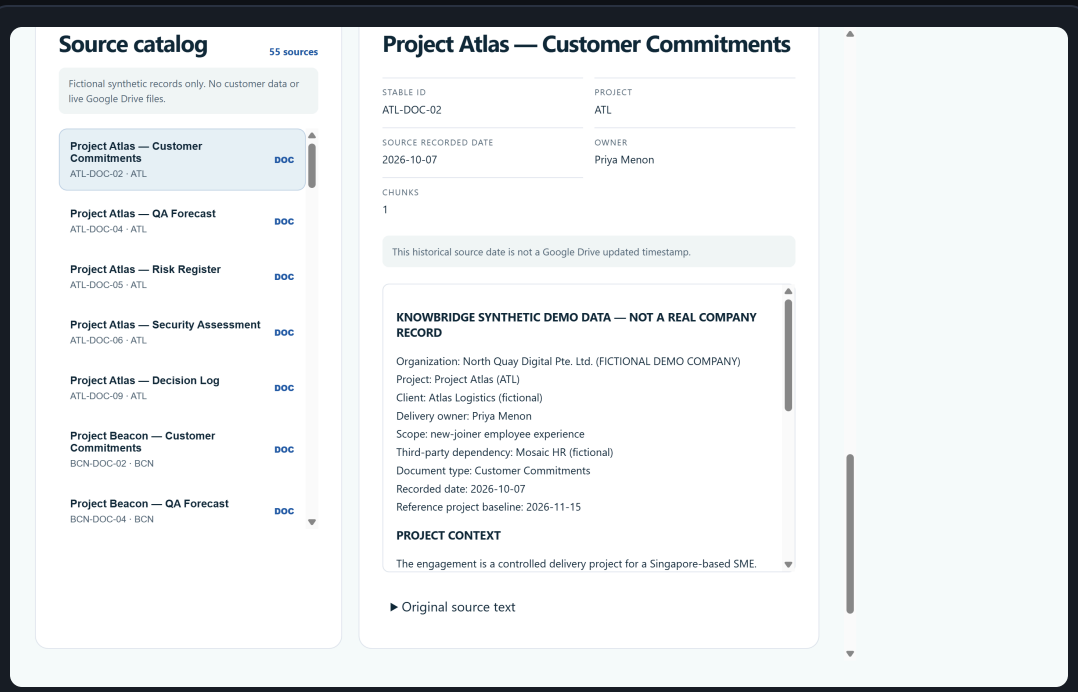
Rough estimates only — all components are usage-based and actual cost scales with query volume. Development uses free-tier / trial credits where available. No commitment or quote is implied.

SNAPSHOTS OF THE PROTOTYPE

Deployed prototype — landing, Ask panel & source catalog



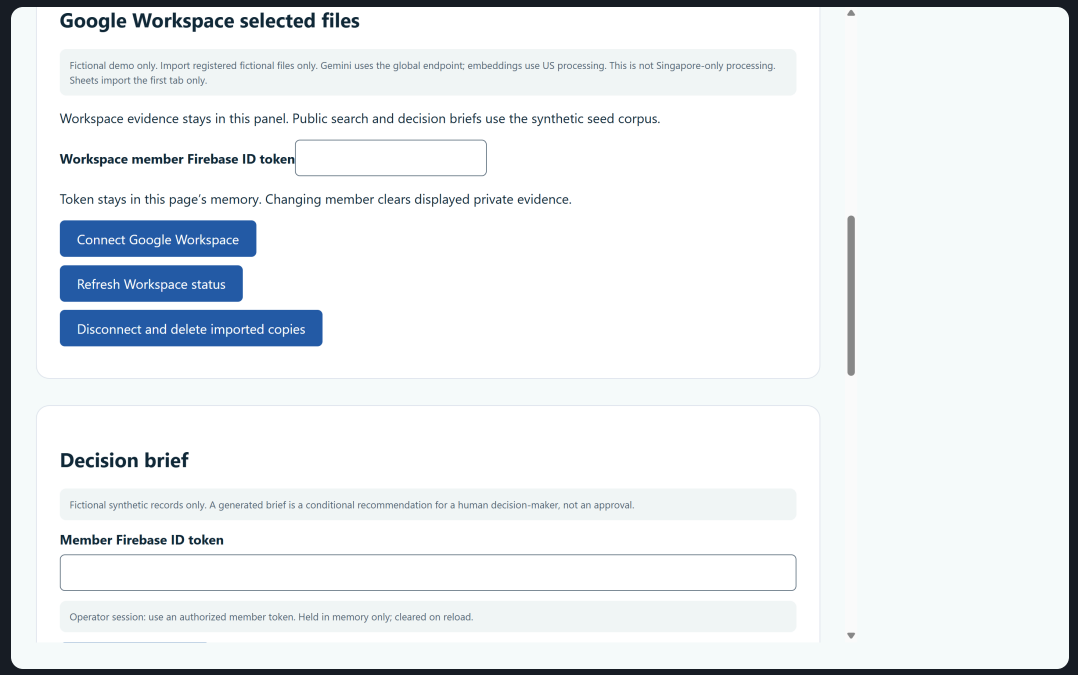
Landing + Ask panel with the Project Harbour question



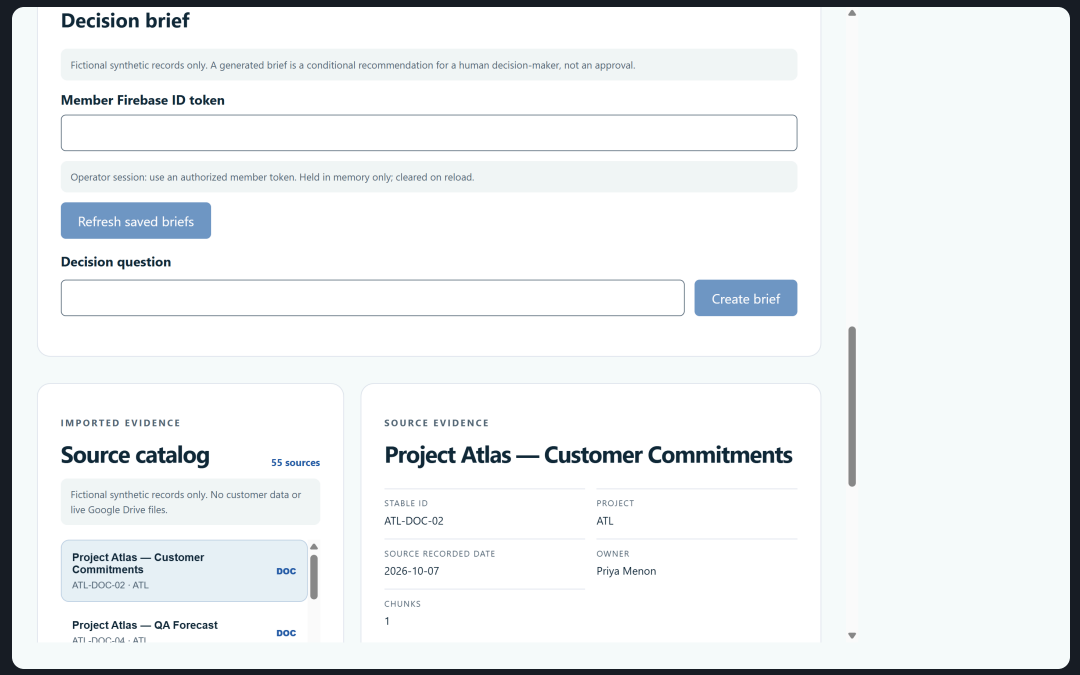
Source catalog + record view: stable ID, recorded date, original text

SNAPSHOTS OF THE PROTOTYPE

Deployed prototype — Workspace import & decision brief



Google Workspace panel: bounded selected-file import + disclosures



Decision brief panel: operator session, human decision-maker

Screenshots are from the live deployed application. All records shown are fictional synthetic data.

PROTOTYPE PERFORMANCE REPORT / BENCHMARKING

Measured results — including failures

168 / 24 / 1,904
fictional Docs / Sheets / rows · 8 projects · 12 gold cases — seed verifier PASS

256 / 256
unit & integration tests PASS · lint, typecheck, build PASS · 8 live semantic adversarial regressions PASS

2 / 12
both final deployed benchmark runs (before/after the reliability fix): every failure was a provider-side service error (HTTP 503), **never an evidence-quality failure**; every pass had fully valid literal citations

9 / 10 / 11 of 12
earlier complete runs (revision knowbridge-00009-7j5; 11/12 local CLI). Totals are never aggregated across runs

Median latency of the pre-fix deployed run including failures: 49,509.5 ms. Full-suite reliability under intermittent provider-side model failures remains an open issue and is reported as such. Artifacts: deployed-evaluation-prefix / deployed-evaluation-postfix (2026-10-17).

ADDITIONAL DETAILS / FUTURE DEVELOPMENT

What's next — and what we will not claim

Roadmap

- **Now:** grounded search, conflict/temporal evidence, decision briefs, save/review, Cloud Run deployment.
- **P1:** Picker selected-file import (built) · lightweight entity linking across records.
- **P2 (post-hackathon):** Gmail / Chat / Calendar connectors, SSO, organization-wide sync.
- **Reliability:** queueing and retries for intermittent model-call 503s — the main open issue.

Honest boundaries

- No productivity guarantee or customer claim — all data is fictional synthetic data.
- Recommendations are conditional; a human verifies every high-impact call.
- Not production-compliant, enterprise-certified or legally approved.
- Model processing is not Singapore-only; real-data import is prohibited.

KnowBridge — Know what your sources actually say.

LINKS

Where to find everything

GITHUB PUBLIC REPOSITORY	**https://github.com/alfianhid/knowbridge**
DEMO VIDEO (3 MINUTES)	**[Demo video link — TBA: recording in progress]**
FINAL PRODUCT LINK	**https://knowbridge-208529907910.asia-southeast1.run.app**

Repository includes the reproducible synthetic dataset (Python seed generator + verifier), architecture notes, evaluation artifacts and run instructions.

All demo companies, people, emails and records are fictional. Theme: Future of Work & Enterprise Productivity. Team KARAKA · AI Builder Cup 2026.